The Spectre Knot

One tile, front and back

Front: the knot tile. Back: what it is. Edge length 41 mm.



The Spectre Knot

A single-strand Celtic knot that fills the plane

Every tile carries the same drawing: pieces of ribbon running from edge to edge, with one internal crossing. Laid without flipping, copies of the tile cover the plane, but never in a repeating pattern. Where tiles meet, their ribbons join up.

The tiles always group into blocks of eight or nine, the blocks into bigger blocks, and so on for ever.

Daedalus Theorem: no closed loops, in any tiling whose blocks grow to cover the whole plane.

Ariadne Theorem: just one endless strand, in one particular tiling grown outwards from a single tile.

The Spectre Knot was devised by Justin Roughley in 2026, working with AI: cortado-j.github.io/SPECTRE_KNOT.

The tile shape is the straight-edged spectre, discovered in 2023 by David Smith, Joseph Samuel Myers, Craig S. Kaplan and Chaim Goodman-Strauss.

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1. Cut round the outer black line, as one piece.
2. Fold along the dashed line, printed sides out.
3. Glue the halves back to back.

Make several and lay them edge to edge: the ribbons meet where the edges touch and cross there at right angles.